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Language Is Not All You Need: Aligning Perception with Language Models

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https://github.com/microsoft/unilm

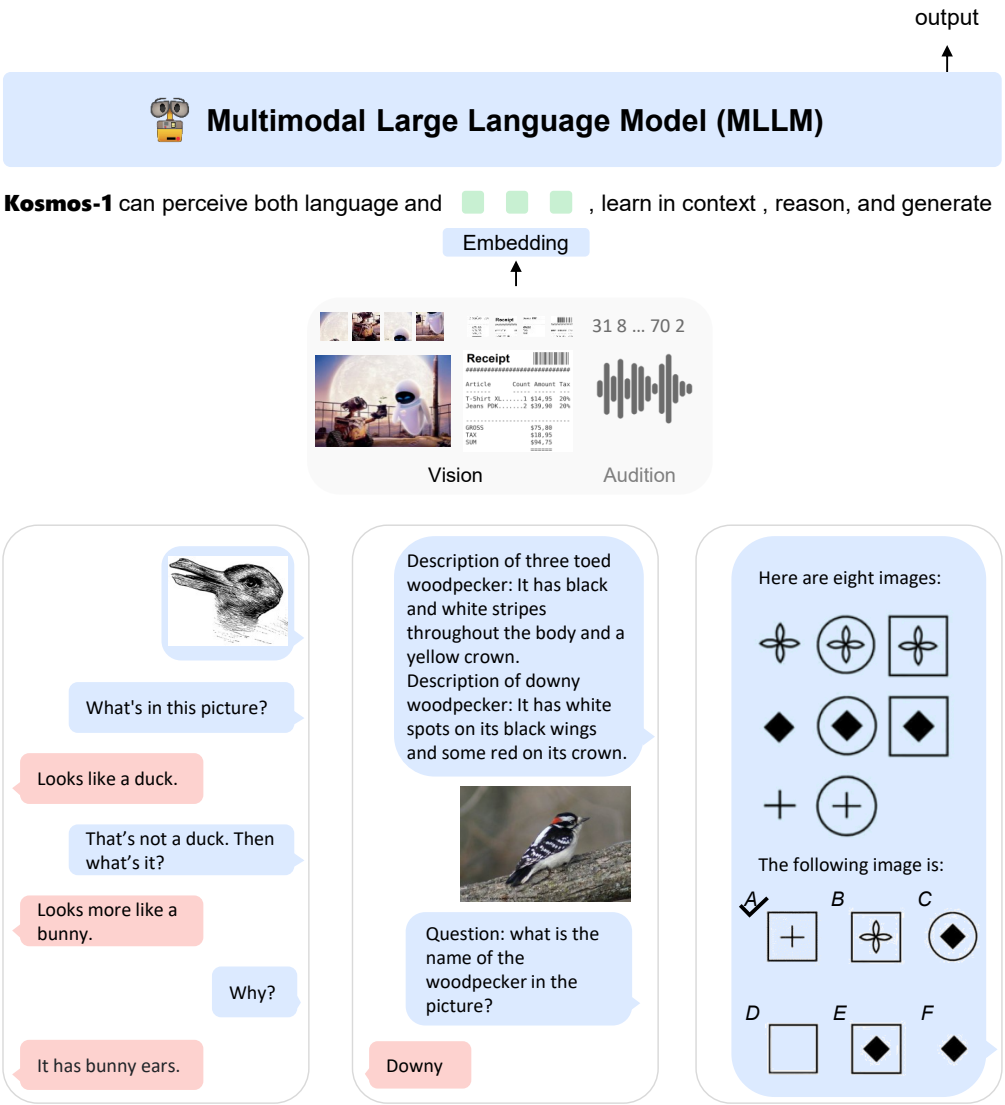

Figure 1: KOSMOS-1 is a multimodal large language model (MLLM) that is capable of perceiving multimodal input, following instructions, and performing in-context learning for not only language tasks but also multimodal tasks. In this work, we align vision with large language models (LLMs), advancing the trend of going from LLMs to MLLMs.

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“The limits of my language means the limits of my world.”

Ludwig Wittgenstein

Abstract

A big convergence of language, multimodal perception, action, and world modeling is a key step toward artificial general intelligence. In this work, we introduce KOSMOS-1[2], a Multimodal Large Language Model (MLLM) that can perceive general modalities, learn in context (i.e., few-shot), and follow instructions (i.e., zero-shot). Specifically, we train KOSMOS-1 from scratch on web-scale multimodal corpora, including arbitrarily interleaved text and images, image-caption pairs, and text data. We evaluate various settings, including zero-shot, few-shot, and multimodal chain-of-thought prompting, on a wide range of tasks without any gradient updates or finetuning. Experimental results show that KOSMOS-1 achieves impressive performance on (i) language understanding, generation, and even OCR-free NLP (directly fed with document images), (ii) perception-language tasks, including multimodal dialogue, image captioning, visual question answering, and (iii) vision tasks, such as image recognition with descriptions (specifying classification via text instructions). We also show that MLLMs can benefit from cross-modal transfer, i.e., transfer knowledge from language to multimodal, and from multimodal to language. In addition, we introduce a dataset of Raven IQ test, which diagnoses the nonverbal reasoning capability of MLLMs.

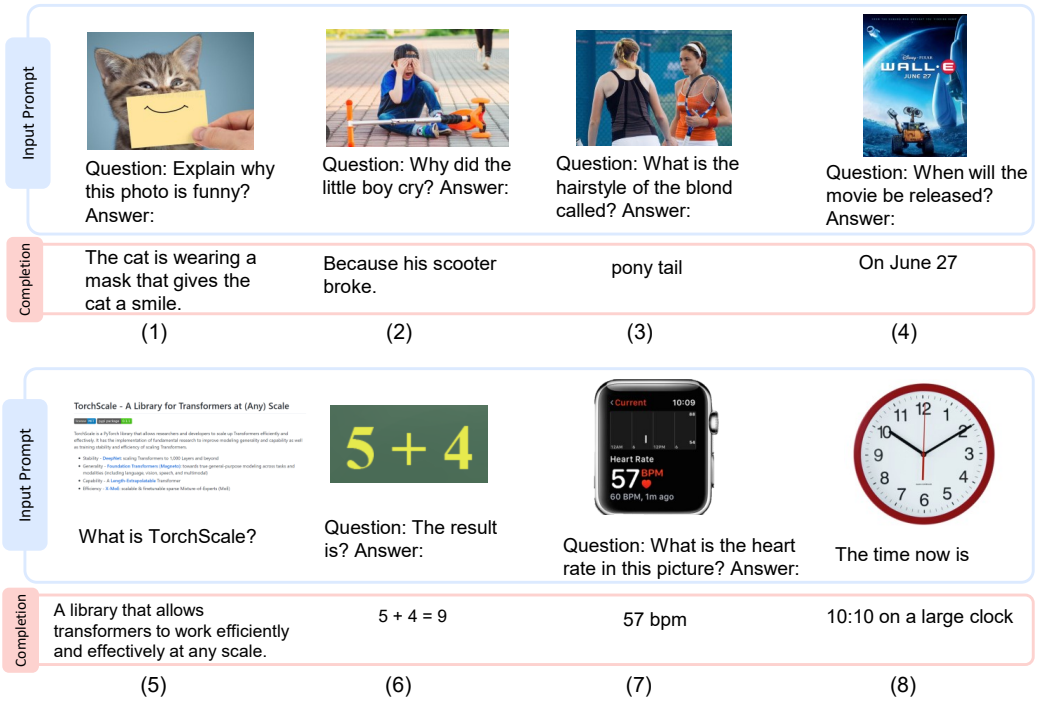

Figure 2: Selected examples generated from KOSMOS-1. Blue boxes are input prompt and pink boxes are KOSMOS-1 output. The examples include (1)-(2) visual explanation, (3)-(4) visual question answering, (5) web page question answering, (6) simple math equation, and (7)-(8) number recognition.

[2]KOSMOS is pronounced as and means “*Cosmos*”.

Figure 3: Selected examples generated from KOSMOS-1. Blue boxes are input prompt and pink boxes are KOSMOS-1 output. The examples include (1)-(2) image captioning, (3)-(6) visual question answering, (7)-(8) OCR, and (9)-(11) visual dialogue.